

# TASK 1 — Dynamic Model of the 6-DOF UAV Quadcopter (FINAL)

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Reference: Mien, T. & Tu, T. (2024), *IJRCS* 4(4), 1712–1730, doi:10.31763/ijrsc.v4i4.1594

**Task 1 as stated:** "Develop the dynamic mathematical model of the 6 DOF UAV Quadcopter in MATLAB and SIMULINK in terms of state space equations."

Status: COMPLETE — 25/25 verification checks passed.

## 1. What the task requires, decomposed

Three deliverables are contained in that one sentence:

- 1. a **dynamic model** — equations describing how the quadcopter moves
- 2. expressed in **state-space form** —  $\dot{x} = Ax + Bu$
- 3. implemented in **MATLAB and Simulink** — both, not one

We add a fourth of our own: **evidence the model is correct**, because an unverified model is an assertion.

## 2. Physical setup

### 2.1 Six degrees of freedom, four actuators

|             | Freedoms      | Symbol                           |
|-------------|---------------|----------------------------------|
| Translation | along x, y, z | $\xi = [X, Y, Z]$                |
| Rotation    | about x, y, z | $\eta = [\varphi, \theta, \psi]$ |

- **$\varphi$  (roll)** — rotation about body x
- **$\theta$  (pitch)** — rotation about body y
- **$\psi$  (yaw)** — rotation about body z

The quadcopter has **6 DOF but only 4 rotors**, so it is **underactuated**. It cannot translate sideways directly — it must **tilt** so that thrust acquires a horizontal component. This appears in the linear model as  $\ddot{x} = g \cdot \theta$  and  $\ddot{y} = -g \cdot \phi$ , and it is the single most important structural fact about the vehicle.

### 2.2 Two coordinate frames

- **Inertial (ground) frame**  $0\text{-}xyz$  — fixed to earth; position and altitude measured here
- **Body frame**  $0_B\text{-}x_B\ y_B\ z_B$  — fixed to the airframe at the centre of gravity; **thrust always acts along  $+z_B$** , whatever the attitude

The entire model arises from reconciling these: forces are produced in the body frame, motion is measured in the inertial frame, and the rotation matrix links them.

### 2.3 The three stated assumptions

Taken verbatim from the problem statement, and each has a modelling consequence:

| Assumption                               | Consequence                                                                   |
|------------------------------------------|-------------------------------------------------------------------------------|
| homogeneous, symmetric block             | inertia tensor is <b>diagonal</b> , $I = \text{diag}(I_{xx}, I_{yy}, I_{zz})$ |
| centre of quadcopter = centre of gravity | no offset terms in the torque balance                                         |
| elasticity neglected, transmission rigid | no flexible-body dynamics                                                     |

### 3. Parameters (Table 1 — given, not chosen)

| Parameter                   | Symbol                            | Value                                     |
|-----------------------------|-----------------------------------|-------------------------------------------|
| Quadcopter mass             | m                                 | 0.516 kg                                  |
| Arm length                  | l                                 | 0.225 m                                   |
| Gravity                     | g                                 | 9.81 m/s <sup>2</sup>                     |
| Inertia moment of the rotor | I_M                               | 3.368×10 <sup>-5</sup> kg·m <sup>2</sup>  |
| Thrust factor of rotor      | k                                 | 2.996×10 <sup>-6</sup> N·s <sup>2</sup>   |
| Drag coefficient            | b                                 | 1.260×10 <sup>-7</sup> N·m·s <sup>2</sup> |
| Inertial constants          | I <sub>xx</sub> , I <sub>yy</sub> | 4.984×10 <sup>-3</sup> kg·m <sup>2</sup>  |
|                             | I <sub>zz</sub>                   | 8.958×10 <sup>-3</sup> kg·m <sup>2</sup>  |

Cross-checked against the source paper's Table 1 — identical in all eight values.

#### Derived quantities

| Quantity                | Formula                  | Value                                |
|-------------------------|--------------------------|--------------------------------------|
| Hover weight            | $W = mg$                 | <b>5.0620 N</b>                      |
| Hover rotor speed       | $\omega = \sqrt{(W/4k)}$ | <b>649.9 rad/s</b> each (~6200 rpm)  |
| Max thrust              | $U1_{\text{max}} = 4W$   | <b>20.248 N</b>                      |
| Max upward acceleration | $U1_{\text{max}}/m - g$  | <b>29.43 m/s<sup>2</sup></b> (≈ 3 g) |

#### A gap in the given data, and how we handled it

The source paper's Eq. (6) includes **air-resistance coefficients Ax, Ay, Az**:

$$\begin{aligned}
m \cdot \ddot{x} + A_x \cdot \dot{x} &= F(C\psi S\theta C\phi + S\psi S\phi) \\
m \cdot \ddot{y} + A_y \cdot \dot{y} &= F(S\psi S\theta C\phi - C\psi S\phi) \\
m \cdot \ddot{z} + mg + A_z \cdot \dot{z} &= F(C\phi C\theta)
\end{aligned}$$

**Neither Table 1 in the problem statement nor Table 1 in the paper gives values for them.** We therefore set  $A_x = A_y = A_z = 0$ , which is the only consistent reading of the data supplied. The code retains the terms ( $P.A_x, P.A_y, P.A_z$  in `quad_params.m`) so a non-zero value can be substituted if a judge asks.

**This choice has a direct consequence in Task 2:** with  $A_z = 0$  the altitude channel is a *pure* double integrator, which is why the Ziegler–Nichols ultimate-gain method is degenerate for us but not for the reference paper.

## 4. Rotor forces to control inputs

### 4.1 Geometry (Fig. 1 — cross / "plus" configuration)

| Rotor | Position             | Spin sense |
|-------|----------------------|------------|
| 1     | (l, 0, 0) — on +x_B  | CW         |
| 2     | (0, -l, 0) — on -y_B | CCW        |
| 3     | (-l, 0, 0) — on -x_B | CW         |
| 4     | (0, l, 0) — on +y_B  | CCW        |

Rotors 1,3 spin opposite to 2,4, so in hover their reaction torques cancel and the vehicle does not spin about z.

### 4.2 Each rotor produces

- **thrust**  $f_i = k \cdot \omega_i^2$  along +z\_B
- **reaction torque**  $\tau_i = b \cdot \omega_i^2$  about z\_B

Both scale with  $\omega^2$  — this is where the input nonlinearity enters.

### 4.3 Deriving the moments with $\tau = r \times F$

For rotor 4 at  $r_4 = (0, l, 0)$  with  $F_4 = (0, 0, f_4)$ :

$$\begin{aligned}
\tau_4 = r_4 \times F_4 &= \begin{vmatrix} i & j & k \\ 0 & l & 0 \\ 0 & 0 & f_4 \end{vmatrix} = (l \cdot f_4, 0, 0) \quad \rightarrow \text{positive roll}
\end{aligned}$$

Repeating for all four rotors gives the **four virtual control inputs**:

|                                                             |                                                    |       |
|-------------------------------------------------------------|----------------------------------------------------|-------|
| $U1 = k(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2)$ | total thrust                                       | [N]   |
| $U2 = k(\omega_4^2 - \omega_2^2)$                           | roll input $\rightarrow \tau_\phi = 1 \cdot U2$    |       |
| $U3 = k(\omega_3^2 - \omega_1^2)$                           | pitch input $\rightarrow \tau_\theta = 1 \cdot U3$ |       |
| $U4 = b(\omega_1^2 + \omega_3^2 - \omega_2^2 - \omega_4^2)$ | yaw torque                                         | [N·m] |

**Why this matters:** controllers are designed against  $U1...U4$  (thrust and three moments), which are physically intuitive and decoupled. Conversion back to individual rotor speeds happens only at the end.

**Verified:** all four rotors at  $\omega_{\text{hover}} = 649.9$  rad/s gives  $U1 = 5.0620$  N (exactly the weight) with all three moments zero — confirming the mapping and the hover point together.

## 5. The nonlinear model

### 5.1 Translational dynamics

Thrust  $F_b = [0, 0, U1]^T$  is rotated into the inertial frame by the ZYX Euler rotation matrix  $R = R_z(\psi)R_y(\theta)R_x(\phi)$ , then Newton's second law with gravity gives:

$$\begin{aligned}\ddot{X} &= (U1/m)(\cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi) \\ \ddot{Y} &= (U1/m)(\cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi) \\ \ddot{Z} &= -g + (U1/m)(\cos \phi \cos \theta)\end{aligned}$$

**The third equation drives all of Task 2.** Vertical acceleration depends on thrust multiplied by  $\cos \phi \cos \theta$  — tilt the vehicle and it loses lift. At  $17^\circ$  of pitch,  $\cos(0.30) = 0.9553$ , so **4.47% of lift disappears**.

### 5.2 Rotational dynamics

Euler's equations with a diagonal inertia tensor, plus the rotor gyroscopic term:

$$\begin{aligned}\ddot{\phi} &= ((I_{yy} - I_{zz})/I_{xx}) \cdot \dot{\theta} \dot{\psi} - (I_M/I_{xx}) \cdot \dot{\theta} \dot{\Omega}_r + (1/I_{xx}) \cdot U2 \\ \ddot{\theta} &= ((I_{zz} - I_{xx})/I_{yy}) \cdot \dot{\phi} \dot{\psi} + (I_M/I_{yy}) \cdot \dot{\phi} \dot{\Omega}_r + (1/I_{yy}) \cdot U3 \\ \ddot{\psi} &= ((I_{xx} - I_{yy})/I_{zz}) \cdot \dot{\phi} \dot{\theta} + (1/I_{zz}) \cdot U4\end{aligned}$$

Three distinct physical effects:

| Term                                                       | Name                     | Meaning                                                                                                     |
|------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------|
| $((I_{yy} - I_{zz})/I_{xx}) \cdot \dot{\theta} \dot{\psi}$ | <b>inertial coupling</b> | rotating about two axes creates torque about the third — the "coupled dynamics" the problem statement cites |
| $(I_M/I_{xx}) \cdot \dot{\theta} \dot{\Omega}_r$           | <b>gyroscopic</b>        | the four spinning rotors resist tilting                                                                     |
| $(1/I_{xx}) \cdot U2$                                      | <b>control</b>           | the commanded moment                                                                                        |

$\Omega_r = \omega_1 - \omega_2 + \omega_3 - \omega_4$  is the **residual rotor speed**. In perfect hover all four are equal so  $\Omega_r = 0$  and the gyroscopic term vanishes; it matters only during manoeuvres.

Note  $I_{zz} \approx 2 \times I_{xx}$ . The vehicle is much harder to rotate about the vertical axis, which is why **yaw is always the slowest channel**.

## 5.3 An approximation you must be able to defend

The equations use **Euler angle rates** ( $\dot{\phi}$ ,  $\dot{\theta}$ ,  $\dot{\psi}$ ) where strictly they should use **body rates** ( $p$ ,  $q$ ,  $r$ ), related by:

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \cos \theta \sin \phi \\ 0 & -\sin \phi & \cos \theta \cos \phi \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

For small angles this matrix  $\rightarrow$  identity. This is the standard simplification used throughout the quadcopter literature, including the cited reference. **It is valid near hover and for moderate angles**, which is our operating regime. State it rather than hide it.

## 6. State-space form

### 6.1 State vector (12 states)

Two states per degree of freedom, since each obeys a second-order ODE:

$$\begin{aligned} \mathbf{x} &= [x \ \dot{x} \ y \ \dot{y} \ z \ \dot{z} \ \phi \ \dot{\phi} \ \theta \ \dot{\theta} \ \psi \ \dot{\psi}]^T \\ \mathbf{u} &= [\delta U_1 \ U_2 \ U_3 \ U_4]^T \end{aligned}$$

### 6.2 Linearisation about hover

Equilibrium: all angles zero, all rates zero,  $U_1 = W = 5.0620 \text{ N}$ . Applying  $\sin \phi \approx \phi$ ,  $\cos \phi \approx 1$  and dropping products of small terms:

| Nonlinear                                               | Linearised                             |
|---------------------------------------------------------|----------------------------------------|
| $\ddot{x} = (U_1/m)(c\phi s\theta c\psi + s\phi s\psi)$ | $\ddot{x} = g \cdot \theta$            |
| $\ddot{y} = (U_1/m)(c\phi s\theta s\psi - s\phi c\psi)$ | $\ddot{y} = -g \cdot \phi$             |
| $\ddot{z} = -g + (U_1/m)c\phi c\theta$                  | $\ddot{z} = \delta U_1/m$              |
| $\ddot{\phi} = \dots + (1/I_{xx})U_2$                   | $\ddot{\phi} = (1/I_{xx}) \cdot U_2$   |
| $\ddot{\theta} = \dots + (1/I_{yy})U_3$                 | $\ddot{\theta} = (1/I_{yy}) \cdot U_3$ |
| $\ddot{\psi} = \dots + (1/I_{zz})U_4$                   | $\ddot{\psi} = (1/I_{zz}) \cdot U_4$   |

### 6.3 The A and B matrices

Non-zero entries only:

|                |                      |                                |
|----------------|----------------------|--------------------------------|
| $A(1,2) = 1$   | $A(2,9) = +9.8100$   | $(\ddot{X} = +g \cdot \theta)$ |
| $A(3,4) = 1$   | $A(4,7) = -9.8100$   | $(\ddot{Y} = -g \cdot \phi)$   |
| $A(5,6) = 1$   | $B(6,1) = 1.9380$    | $(= 1/m)$                      |
| $A(7,8) = 1$   | $B(8,2) = 45.1445$   | $(= 1/I_{xx})$                 |
| $A(9,10) = 1$  | $B(10,3) = 45.1445$  | $(= 1/I_{yy})$                 |
| $A(11,12) = 1$ | $B(12,4) = 111.6321$ | $(= 1/I_{zz})$                 |

$C = I_{12}, D = 0$ .

$A(2,9)$  and  $A(4,7)$  are underactuation made visible. They say the only way to accelerate horizontally is to tilt — there is no  $B$  entry that translates directly.

## 6.4 Four decoupled channels

| Channel  | Equation                       | Transfer function        | Numeric                 |
|----------|--------------------------------|--------------------------|-------------------------|
| Altitude | $\ddot{Z} = \delta U1/m$       | $1/(m \text{ s}^2)$      | $1/(0.516 \text{ s}^2)$ |
| Roll     | $\ddot{\phi} = (1/I_{xx})U2$   | $1/(I_{xx} \text{ s}^2)$ | $45.14/\text{s}^2$      |
| Pitch    | $\ddot{\theta} = (1/I_{yy})U3$ | $1/(I_{yy} \text{ s}^2)$ | $45.14/\text{s}^2$      |
| Yaw      | $\ddot{\psi} = (1/I_{zz})U4$   | $1/(I_{zz} \text{ s}^2)$ | $111.63/\text{s}^2$     |

**Every channel is a double integrator.** This is the most useful structural fact in the whole problem and drives every decision in Task 2.

## 7. Why "double integrator" matters

$G(s) = 1/(m \text{ s}^2)$  has **two poles at the origin**:

1. **Marginally stable, not stable.** Constant force  $\rightarrow$  output grows as  $t^2$ . Verified: a 2% thrust error diverges quadratically.
2. **Proportional control alone cannot stabilise it.**  $m \cdot s^2 + K_p = 0$  gives poles at  $\pm j\sqrt{K_p/m}$  — sustained oscillation at every gain. **Derivative action is mandatory.**
3. **The ZN ultimate-gain method is degenerate** — no unique  $K_u$  exists. (See TASK2\_final §3 for the important qualification regarding the reference paper.)

## 8. Verification — the evidence

Anyone can write equations; the mark is in proving them. All checks pass:

| # | Check                               | Result                    | Interpretation                       |
|---|-------------------------------------|---------------------------|--------------------------------------|
| 1 | Table 1 parameters (9 values)       | all exact                 | inputs correct                       |
| 2 | Hover residual $\ \dot{x}\ $        | <b>0.000e+00</b>          | hover is an exact equilibrium        |
| 3 | <code>max\ A - A_numerical\ </code> | <b>1.636e-12</b>          | hand derivation = finite differences |
| 4 | <code>max\ B - B_numerical\ </code> | <b>3.122e-10</b>          | same for B                           |
| 5 | Structural entries (6 checks)       | all match                 | physics is in the right slots        |
| 6 | <code>rank(ctrb(A,B))</code>        | <b>12 of 12</b>           | fully controllable                   |
| 7 | Open-loop poles                     | all 12 at origin          | cannot fly without feedback          |
| 8 | Rotor mapping at hover              | U1 = 5.0620 N, moments 0  | mapping and hover consistent         |
| 9 | Roll/pitch sign convention          | both positive as expected | no inverted axis                     |

**Check 3 is the one to put on a slide.** It finite-differences the nonlinear model and compares against the hand-derived matrices entry by entry. Agreement to twelve decimal places converts "trust our algebra" into "here is the proof."

Run it yourself: `verify_all` in the `solution/` folder.

## 9. Files — what each one does

### Core model files

| File                         | Purpose                                                                                                                                                                                                                                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>quad_params.m</code>   | Returns struct <code>P</code> with all 8 Table 1 parameters, derived quantities ( $W$ , $\omega_{\text{hover}}$ , $U1_{\text{max}}$ ), actuator limits, drag placeholders ( $A_x, A_y, A_z = 0$ ), and the 100 Hz controller sample time.<br><b>Every other file gets its numbers from here</b> — nothing is hard-coded. |
| <code>quad_dynamics.m</code> | The full nonlinear 6-DOF model. Input: state <code>x</code> (12×1), control <code>U</code> (U1–U4 plus $\Omega_r$ ), params <code>P</code> , optional disturbance force. Output: $\dot{x}$ (12×1). Implements §5.1 and §5.2 exactly.                                                                                     |
| <code>rotor2U.m</code>       | Maps four rotor speeds <code>[w1 w2 w3 w4]</code> to <code>[U1 U2 U3 U4]</code> and the residual speed $\Omega_r$ , using the $\tau = r \times F$ geometry of §4. Sign conventions documented inline.                                                                                                                    |

## Task 1 scripts

| File                            | Purpose                                                                                                                                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>task1_statespace.m</code> | The Task 1 deliverable. Builds A, B, C, D analytically; verifies by finite-differencing the nonlinear model; forms the four channel transfer functions; runs the controllability test; simulates open-loop divergence. Saves <code>task1_model.mat</code> . |
| <code>verify_all.m</code>       | Independent re-verification. Reloads saved results and re-derives them from scratch rather than trusting the run that produced them. 25 checks. <b>Run this before the demo.</b>                                                                            |
| <code>smoke_test.m</code>       | Fast 10-point regression check of the whole package — toolbox availability, hover equilibrium, rotor signs, closed-loop tracking, <code>pidtune</code> , <code>fitnet</code> . Seconds to run; use it after any edit.                                       |

## Outputs

| File                                                  | Contents                                             |
|-------------------------------------------------------|------------------------------------------------------|
| <code>task1_model.mat</code>                          | <code>P, A, B, C, D, sys_full, sys_alt, G_alt</code> |
| <code>figures/01_task1_openloop_divergence.png</code> | 2% thrust error diverging as $t^2$                   |
| <code>figures/02_task1_four_channels_pzmap.png</code> | pole-zero maps of all four channels                  |

## How to run

```
cd 'C:\Users\LENOVO\Downloads\HACKSIMU8\solution'
task1_statespace      % builds and verifies the model
verify_all           % independent 25-check verification
```

## 10. Presenting Task 1 (about 3 minutes)

1. **"6 DOF, 4 actuators — underactuated."** Show the state vector; point at  $\ddot{\mathbf{x}} = \mathbf{g} \cdot \boldsymbol{\theta}$  and say the quadcopter must tilt to translate.
2. **"Thrust and torques come from  $\omega^2$ ."** Show the U1–U4 mapping; mention  $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ .
3. **"Here are the six nonlinear equations."** Point at `cos  $\phi$  cos  $\theta$`  in the altitude equation and flag that it drives Task 2.
4. **"Linearised about hover it decouples into four double integrators."** Show the table.
5. **"And here is the proof it's right."** `max|A - A_num| = 1.6e-12`, rank 12, zero hover residual.
6. **"All twelve poles at the origin — it can't fly open loop."** → hand to Task 2.

## Questions and answers

**"Why 12 states?"** Six degrees of freedom, each second-order, so two states each: position and velocity.

**"Why linearise if you have the nonlinear model?"** Bode, root locus, margins and `pidtune` all assume linearity. We keep the nonlinear model for *simulation* and use the linear one for *design*, then verify the



design back on the nonlinear model — Task 2's Method D does exactly that.

**"Is the linear model valid at large angles?"** No, it assumes small angles. That is precisely why the altitude controller uses feedback linearisation instead, which stays exact at any tilt (verified to 0.000262 m at 25.8°).

**"You used Euler rates rather than body rates."** Correct and deliberate — they coincide for small angles, and it is the formulation used by the cited reference. Here is the exact transformation between them.

**"What did you neglect?"** Aerodynamic drag on the airframe (coded but zero, because Table 1 gives no values — the paper's Eq. 6 has the terms but its Table 1 omits them too), blade flapping, motor dynamics, ground effect, and airframe flexibility (excluded by the stated assumptions).

**"How do you know the model is right?"** Twenty-five independent checks, including a finite-difference re-derivation of A and B that agrees to 1.6e-12. Run `verify_all` and watch.